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United States Patent	12410650
Kind Code	B2
Date of Patent	September 09, 2025
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Swing door operator

Abstract

A door operator including an output shaft, a ball screw, a powered driver, and a yoke. The output shaft is configured to be coupled to a door. The powered driver is operatively coupled to the ball screw and configured to rotate the ball screw in a first rotational direction. The yoke is coupled to the ball screw and configured to translate in response to rotation of the ball screw. The yoke is configured to rotate the output shaft.

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Appl. No.:	17/681233
Filed:	February 25, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230272661 A1	Aug. 31, 2023

Publication Classification

Int. Cl.: E05F15/63 (20150101); E05F1/10 (20060101); E05F15/53 (20150101)

U.S. Cl.:

CPC E05F15/63 (20150115); E05F1/105 (20130101); E05F15/53 (20150115); E05F2015/631 (20150115); E05Y2201/11 (20130101); E05Y2201/499 (20240501); E05Y2600/56 (20130101); E05Y2800/70 (20130101); E05Y2900/132 (20130101)

Field of Classification Search

CPC: E05F (2015/631); E05Y (2201/11); E05Y (2201/492); E05Y (2201/702); E05Y (2201/224); E05Y (2201/638); E05Y (2201/682); E05Y (2201/688); E05Y (2201/70); E05Y (2600/56); E05Y (2600/12); E05Y (2800/70); E05Y (2800/174); E05Y (2800/296)

USPC: 49/340; 49/341; 49/342; 49/344; 49/345; 49/346

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Background/Summary

TECHNICAL FIELD

(1) This disclosure is generally directed to an operator for a door, specifically, a swing door. In some examples, a ball screw driven by a motor moves a yoke which causes rotation of an output shaft to open the door and also compresses biasing springs configured to return the door to a closed position.

BACKGROUND

(2) Various operators for automatically opening and/or closing doors are known, particularly with respect to swing doors which swing open by pivoting around hinges mounted to a door frame. Some operators may use an electric motor, for example, to both open and close the door by rotating a spindle which is connected to an arm that is, in turn, connected to the door. Rotating the spindle causes displacement of the arm which causes the door to transition between open and closed conditions. Some operators use a motor for opening the door and use a non-powered mechanism for closing the door such as a spring which biases the door toward the closed position.

(3) Some known door operators may include a large number of moving parts including gears, spindles, levers, and other components. During use, some of these components may be subjected to backlash due to design tolerances and other considerations. Backlash may lead to a considerable error in positioning within the components and, in turn, the door.

(4) Some door operators may be configured for a specific installation orientation. That is, some door operators are limited to use as a left-hand operator or a right-hand operator and the appropriate operator must be selected for a given door configuration. In this regard, a left-hand operator may be unsuitable for use in a right-hand setup and vice versa.

(5) A need accordingly exists for a swing door operator that addresses one or more shortcomings of conventional swing door operators.

SUMMARY

(6) The present disclosure is directed to a swing door operator that may address one or more of the challenges found in conventional swing door operators. Some implementations may include a ball screw that is driven by a motor to open the door and is driven by springs to close the door.

(7) According to some examples, the present disclosure is directed to a door operator that may include an output shaft, a ball screw, a powered driver, and a yoke. The output shaft may be configured to be coupled to a door. The powered driver may be operatively coupled to the ball screw and configured to rotate the ball screw in a first rotational direction. The yoke may be coupled to the ball screw and configured to translate in response to rotation of the ball screw. The yoke may be configured to rotate the output shaft.

(8) In some examples, a door operator may include a cam assembly coupled to the yoke. The cam assembly may be configured to rotate about a rotation axis of the output shaft during translation of the yoke. The rotation axis of the output shaft may be substantially transverse to a rotation axis of the ball screw. A yoke may include an elongated slot having a longitudinal axis oriented transverse

to the rotation axis of the ball screw. A cam assembly may extend into the elongated slot. A cam assembly may include a guide member configured to roll along a surface defining the elongated slot during translation of the yoke. A door operator may include a directional block to prevent the guide member from crossing the longitudinal axis of the door operator.

(9) In some examples, a door operator may include at least one pin oriented transverse to a longitudinal axis of the ball screw. The at least one pin may be configured to extend through at least one slot in the yoke. The at least one slot in the yoke may be oriented parallel to the longitudinal axis of the ball screw.

(10) In some examples, a barrier operator may include at least one spring configured to rotate the ball screw in a second rotational direction opposite the first rotational direction. A ball screw nut may be disposed about the ball screw and coupled to the yoke. The ball screw nut may be configured to pull the yoke in a first linear direction when the ball screw is rotated in the first rotational direction and to push the yoke in a second linear direction when the ball screw is rotated in the second rotational direction. At least one rod may extend longitudinally within each spring of the at least one spring. The yoke may include at least one rod recess configured to receive an end of the at least one rod as the yoke is translated. The yoke may be configured to compress the at least one spring as the yoke translates in response to rotation of the ball screw in the first rotational direction. The at least one spring may be configured to expand and push the yoke causing rotation of the ball screw in the second rotational direction.

(11) In some examples, a door operator has a first installation configuration in which operation of the powered driver causes rotation of the output shaft in a third rotational direction and a second installation configuration in which operation of the powered driver causes rotation of the output shaft in a fourth rotational direction opposite the third rotational direction.

(12) According to some examples, the present disclosure is directed to a system including a door and a door operator. The door operator may include an output shaft, a ball screw, a powered driver, and a yoke. The output shaft may be configured to be coupled to the door. The powered driver may be operatively coupled to the ball screw and configured to rotate the ball screw in a first rotational direction. The yoke may be coupled to the ball screw and configured to translate in response to rotation of the ball screw. The yoke may be configured to rotate the output shaft.

(13) In some examples, the door operator may include at least one spring configured to rotate the ball screw in a second rotational direction opposite the first rotational direction.

(14) In some examples, a system may include one or more of a swing arm coupling the door operator and the door, a frame hingedly coupled to the door and supporting the door operator, a track mounted to the door and slidably supporting an end of the swing arm, a bracket mounted to the door and pivotally supporting an end of the swing arm, or an actuator configured to trigger actuation of the powered driver.

(15) According to some examples, the present disclosure is directed to a method for operating a door operator. The method may include rotating a ball screw in a first rotational direction with a powered driver operatively coupled to the ball screw, translating a yoke coupled to the ball screw using the rotation of the ball screw, and rotating an output shaft using the translation of the yoke, the output shaft couplable to a door.

(16) In some examples, a method may include compressing, during the translation of the yoke, the at least one spring, and rotating the ball screw in a second rotational direction, opposite the first rotational direction, by expanding the at least one spring. Rotating the output shaft using the translation of the yoke may include rotating a cam assembly engaged with the yoke. The cam assembly may include the output shaft.

(17) It is to be understood that both the foregoing general description and the following drawings and detailed description are exemplary and explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. In that regard, additional aspects, features, and advantages of the present disclosure will be apparent to one

skilled in the art from the following. One or more features of any embodiment or aspect may be combinable with one or more features of other embodiment or aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings illustrate implementations of the systems, devices, and methods disclosed herein and together with the description, serve to explain the principles of the present disclosure.
- (2) FIG. 1 is a front view of an embodiment of a system including a door operator according to an example implementation of the present disclosure.
- (3) FIG. 2 is a perspective view of a portion of the system of FIG. 1.
- (4) FIG. 3 is a bottom view of the system of FIG. 1, showing the door in an open configuration in dashed lines and a closed configuration in solid lines.
- (5) FIG. 4 is a side view of an embodiment of a door operator according to an example implementation of the present disclosure.
- (6) FIG. 5 is a bottom view of the door operator of FIG. 4 in a closed-door configuration with the housing cover removed.
- (7) FIG. 6 is a bottom view of the door operator of FIG. 5 with a cover plate removed.
- (8) FIG. 7 is a side view of the door operator of FIG. 5.
- (9) FIG. 8 is a bottom view of the door operator of FIG. 5 in an open-door configuration.
- (10) FIG. 9 is a perspective view of a yoke according to an example implementation of a door operator of the present disclosure.
- (11) FIG. 10 is a perspective view of a cam assembly according to an example implementation of a door operator of the present disclosure.
- (12) FIG. 11 provides a schematic illustration of an embodiment of a system for opening a door according to an example implementation of the present disclosure.
- (13) FIG. 12 provides a schematic diagram of an embodiment of a door system according to the present disclosure.
- (14) FIG. 13 is a flowchart of a method for operating a door operator according to an example implementation of the present disclosure.
- (15) These Figures will be better understood by reference to the following Detailed Description.

DETAILED DESCRIPTION

- (16) For the purpose of promoting an understanding of the principles of the present disclosure, reference will now be made to the implementations illustrated in the drawings and specific language will be used to describe them. It will nevertheless be understood that no limitation of the scope of the disclosure is intended. Any alterations and further modifications to the described devices, instruments, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In addition, this disclosure describes some elements or features in detail with respect to one or more implementations or Figures, when those same elements or features appear in subsequent Figures, without such a high level of detail. It is fully contemplated that the features, components, and/or steps described with respect to one or more implementations or Figures may be combined with the features, components, and/or steps described with respect to other implementations or Figures of the present disclosure. For simplicity, in some instances the same or similar reference numbers are used throughout the drawings to refer to the same or like parts.
- (17) The present disclosure is directed to a swing door operator that may include a motorized ball screw driving rotation of a cam and, in turn, an output shaft. Motor driven movement of the ball screw also compresses one or more springs that reverse movement of the operator components

upon stopping of the motor. A door operator of the present disclosure may be easily transitioned from a right-hand unit to a left-hand unit by adjustment and/or repositioning of one or more components, and in some implementations, may have little or no backlash.

(18) FIGS. **1-3** illustrate an embodiment of a system **90** which includes a swing door operator **100** and a door **102** according to the present disclosure. Door operator **100** is typically mounted above the door **102**. For example, it may be mounted to the door frame, to a header above the door frame, to a wall above the door frame, or even within a wall. It is also contemplated that a door operator may be positioned along a side of or adjacent to a door. A swing arm assembly **104** may form a part of the door operator **100** or may be configured to cooperate with the door operator **100**. Swing arm assembly **104** is attached to the door operator **100** at one end and is attached to the door **102** at the other end. Depending on the implementation, the swing arm assembly **104** may include a single member or link or may be comprised of a plurality of members or links. In the embodiment illustrated in FIG. **1**, the swing arm assembly **104** is attached to the door **102** via a track **106**. The track **106** also may form a part of the door operator **100** or may be separate from the door operator **100**. An end of the swing arm assembly **104** may slide along the track **106** as the door **102** opens and closes.

(19) In FIG. **2**, it can be seen that swing arm assembly **104** in this implementation comprises first arm member **104a** and second arm member **104b** which are pivotally connected at a joint. In this illustration, second arm member **104b** is attached to the door **102** via a bracket **108**, in lieu of a track, and pivots with respect thereto as the door **102** opens and closes. The swing arm assembly **104** connects to the door operator **100** at a spindle **110** as shown in FIG. **3**.

(20) FIG. **3** illustrates how the first arm member **104a** and second arm member **104b** of the swing arm assembly **104** interact to transition from a closed configuration of the door **102** to an open configuration of the door **102'** as it pivots about a hinge and as spindle **110** rotates the first arm member **104a**.

(21) FIGS. **4-11** illustrate an embodiment of a door operator **100** according to the present disclosure. As shown in FIG. **4**, spindle **110** may have a notched or toothed surface that interfaces with a corresponding surface of first arm member **104a**. Alternatively, first arm member **104a** may be permanently attached to spindle **110** or may be screwed or otherwise fastened thereto. A housing cover **120** may enclose a space housing internal components of the door operator **100**. The housing cover **120** may improve an aesthetic appearance of the door operator **100** and prevent injury that may otherwise be caused by the movement of the internal components. A motor **114** may be secured to a side or end of the door operator **100**. The motor **114** may be disposed in any orientation or arrangement which facilitates powering the door operator **100**. Although illustrated and described as an electric motor, motor **114** may comprise any suitable mechanism for powering the drive unit **128** of the door operator **100**, such as a pneumatic or hydraulic piston. In this regard, the motor **114** may be referred to herein as a “driver” or “powered driver” which encompasses any suitable means for driving operation of the door operator **100**. In the illustrated example, the motor **114** extends outside the housing cover **120**. However, the motor **114** may be concealed within the housing cover **120** in some examples. Housing cover **120** may be secured to the door operator **100** with any suitable fastener or latch or, in some examples, may be snap fit to a primary bracket **123** or other component of the door operator **100**.

(22) Turning to FIGS. **5-7**, door operator **100** is illustrated in a closed-door configuration. In other words, these figures show the internal components of the door operator **100** in a position corresponding to the door being closed. The drive unit **128**, as that term is used herein, encompasses the moving parts of the door operator **100** utilized to open the door including, but not necessarily limited to, the ball screw **116**, the ball screw nut **118**, the yoke **122**, the cam assembly **136**, and the spindle **110**. A number of transverse brackets are mounted to the primary bracket **123** to support the various internal components of the door operator **100**. A motor mounting bracket **124** is secured to the primary bracket **123** near one end and an end bracket **125** is secured to primary

bracket **123** near the other end. A bearing bracket **134** is disposed between the motor mounting bracket **124** and the end bracket **125**. The bearing bracket **134** supports a bearing, such as a ball bearing or other conventional bearing. A number of transverse plates may also be secured to the primary bracket **123**, including a first cover plate **112a** and a second cover plate **112b**. One or more of the motor mounting bracket **124**, the end bracket **125**, the bearing bracket **134**, the first cover plate **112a**, and the second cover plate **112b** may be fastened to the primary bracket **123** using bolts, screws, or any other suitable fastener. One or more of these components may alternatively be welded or otherwise permanently affixed to the primary bracket **123**.

(23) A ball screw **116** with a helical channel along a portion of its length extends between the motor **114** at a proximal end and a yoke **122** at a distal end. Details of the yoke **122** are provided in FIG. **9** and the discussion related thereto below. Although the helical channel of a ball screw is typically smooth and rounded as compared to a threaded shaft of a typical bolt or screw, the portion of the ball screw **116** including the helical channel may be referred to herein as “threaded.” A coupler **126** is configured to secure a drive shaft of the motor **114** to the proximal end of the ball screw **116**. The coupler **126** is illustrated as being positioned between the motor mounting bracket **124** and the bearing bracket **134**. However, it should be appreciated that the coupler **126** may be positioned proximally of the motor mounting bracket **124** or distally of the bearing bracket **134** depending on the configuration of the motor **114** being used, particularly the length of its drive shaft. The proximal end of the ball screw **116**, to which the coupler **126** is secured, may be unthreaded.

(24) A ball screw nut **118** is secured to the yoke **122**, for example by one or more fasteners, and contains one or more balls positioned between the helical channel of the ball screw **116** and an internal surface of the ball screw nut **118**. The balls inside the ball screw nut **118** may be steel ball bearings or any suitable similar structure. The ball screw nut **118** may include one or more internal or external return channels to circulate the balls within the ball screw nut **118** during translation of the ball screw nut **118** along the ball screw **116**. Ball screw nuts of this type are generally known to those of ordinary skill in the art. Some examples of ball screw nuts are provided in U.S. Pat. Pub. No. 2012/0325036 entitled “Ball Screw and Manufacturing Method of Nut for Ball Screw,” U.S. Pat. Pub. No. 2005/0204836 entitled “Seal for Ball Screw and Ball Screw Utilizing Same,” and U.S. Pat. No. 3,720,116 entitled “Arrangement for Preloading Ball Screw Assemblies & Method of Manufacture of the Ball Screw Nut Therefor,” all of which are incorporated by reference herein in their entireties.

(25) A bearing positioned within the bearing bracket **134** may support an unthreaded proximal portion (or “journal”) of the ball screw **116** and keep it positioned in line with the rotation axis of the motor **114**, which lies along the longitudinal axis **129** of the door operator **100**. In some examples, a portion of the journal or a portion of the shaft of the ball screw **116** proximal of the journal may be threaded to receive a locking nut. The locking nut may be positioned between the coupler **126** and the bearing plate **134** and may prevent the ball screw **116** from being pulled distally toward the yoke **122** and away from the coupler **126** during operation. That is, the locking nut may have an outer diameter greater than a diameter of an opening in the bearing plate **134** through which the ball screw **116** extends. In this regard, in the event that a pulling force is exerted on the ball screw **116** during operation that is sufficiently strong to tend to pull the ball screw **116** out of the coupler **126**, the locking nut may contact the bearing plate **134** and prevent such movement.

(26) The distal end of the ball screw **116** is supported and maintained in alignment with the rotation axis by the yoke **122** and/or ball screw nut **118**. While illustrated with the rotation axis of the motor drive shaft and ball screw **116** coinciding with the longitudinal axis **129** of the door operator **100**, it will be appreciated that the motor **114** and ball screw **116** may be shifted in any direction transverse to the longitudinal axis **129**. In this regard, it should be appreciated that, in some examples, the longitudinal axis **129** of the door operator **100** may be defined as intersecting the axis of rotation of the spindle **110** and extending parallel to an axis of translation of the yoke **122**.

(27) During rotation of the drive shaft of the motor **114**, the coupler **126** causes rotation of the ball screw **116** which rotates within the bearing in the bearing bracket **134** and the ball screw nut **118**. Because the motor **114** and ball screw **116** are positionally fixed, this rotation causes movement of the ball screw nut **118** toward the motor **114**. Movement of the ball screw nut **118** pulls the yoke **122** in the same direction. The yoke **122** slides longitudinally along two pins **111a**, **111b** extending between a first cover plate **112a** and a second cover plate **112b**. While the pins **111a**, **111b** remained fixed in position by the cover plates **112a**, **112b**, the pins slide relative to the yoke **122** within first and second respective slots **149a**, **149b**. The pins **111** assist in keeping the yoke **122** positioned along the longitudinal axis **129**. Spacer strips **121** may be positioned within the door operator **100** between the first and second cover plates **112a**, **112b** and the yoke **122**. These spacer strips **121** may be formed from a polymer, a plastic, a metal, or any other suitable material to support the yoke **122** and keep it aligned with the longitudinal axis **129**. Four spacer strips **121** are provided in the illustrated example, two on one side of the yoke **122** and two on the other side of the yoke, each of the two spacer strips **121** on one side of the yoke spaced forward and rearward (top and bottom in the illustrated orientation of FIG. 6) across a surface of the yoke. The spacer strips **121** may be fixed to the yoke **122** or to a respective cover plate **112a**, **112b**. It will be appreciated that any suitable number of spacer strips, **121**, pins **111**, and slots **149** may be used without departing from the scope of the present disclosure.

(28) As the yoke **122** translates linearly along the longitudinal axis **129**, it pulls on a free end of a cam assembly **136** causing rotation thereof. Details of the cam assembly **136** are provided in FIG. 10 and the discussion related thereto below. An end of the cam assembly **136** opposite the free end is positionally fixed about a rotation axis (axis **156** of FIG. 10) that is approximately transverse to the rotation axis of the motor **114** such that movement of the yoke **122** along the longitudinal axis **129** causes rotation of the cam assembly **136** about rotation axis **156** of the cam assembly. This rotation of the cam assembly **136** drives rotation of the spindle **110** to operate the swing arm assembly **104** as described above.

(29) Two springs **130a**, **130b** extend between the motor mounting bracket **124** and the yoke **122**. Any number of springs, including less than or more than two springs, may be used without departing from the scope of the present disclosure. The springs **130a**, **130b** may each be positioned over a respective rod **132a**, **132b**. The rods **132a**, **132b** may be secured to the motor mounting bracket **124** at a proximal end. A distal end of each rod **132a**, **132b** may be received in and slide within a rod aperture formed in the yoke **122**. In this regard, the rods **132a**, **132b** may help retain the springs **130a**, **130b** in their respective positions. During translation of the yoke **122** caused by the motor **114**, the springs **130a**, **130b** are compressed between the yoke **122** and the motor mounting bracket **124** as the drive unit **128** transitions from the closed-door configuration shown in FIGS. 5-7 to an open-door configuration shown in FIG. 8. In this open-door configuration of FIG. 8, the yoke **122** as been translated toward the motor **114** (to the right in the illustration) resulting in rotation of the cam assembly **136** and compression of the springs **130a**, **130b**. When the motor **114** is powered off, disengaged from the drive unit **128**, or otherwise ceases powering the drive unit **128**, the potential energy stored in the compressed springs **130a**, **130b** is converted into kinetic energy as the springs expand and push the yoke **122** away from the motor **114** which, in turn, rotates the cam assembly **136** and the spindle **110** as the springs return the drive unit **128** back to the closed-door configuration. In an alternative example of a door operator of the present disclosure, the springs **130** may be configured to operate in tension rather than compression. In such an embodiment, a spring **130** may be secured to the yoke **122** at one end and to a cover plate **112a**, **112b** or other component distal of the yoke at the other end. In this regard, movement of the yoke **122** toward the motor **114** during operation of the motor increases tension in the spring **130** which can be released upon stopping the motor **114**. In the closed-door configuration, the springs **130a**, **130b** may be in a neutral (no compression or tension) position or may be in a state of pre-compression or pre-tension to bias the door toward a closed position. It should be appreciated that

the springs **130a**, **130b** may extend from the yoke **122** only to the bearing bracket **134** or to an additional bracket (not shown) positioned distally of the motor mounting bracket **124**.

(30) A directional block **127** may be threadedly engaged with the end bracket **125** to advance or retract with respect thereto. A head of the directional block **127** may be rotated by hand or by a tool (e.g., screwdriver or hex key) inserted into a tool engagement feature (e.g., recess). An end or tip of the directional block **127** opposite the head extends through the end bracket **125** to contact the yoke **122**, thereby preventing the yoke from travelling far enough to allow the cam assembly **136** to cross over a longitudinal axis **129** of the door operator **100**. The directional block **127** may ensure the direction of operation of the door operator **100** remains consistent with its installation configuration, e.g., left-hand or right-hand operation. In that regard, the directional block **127** prevents the free end of the cam assembly **136** from crossing the longitudinal axis **129** when the drive unit **128** is in the closed-door configuration. To transition the door operator **100** to a different installation configuration, the directional block **127** may be retracted, the cam assembly **136** may be moved to the other side of the yoke **122** (i.e., across the axis **129**), and the directional block may be extended back into contact with the yoke **122**. The spindle **110** may be moved to an opposing side of the cam assembly **136**.

(31) FIG. **9** illustrates an example of a yoke **122** including a yoke body **143** and a flange **142** disposed at one end of the yoke body **143**. The flange **142** may be configured to contact the cover plates **112a**, **112b** when the drive unit **128** is in the closed-door configuration. The flange **142** may include a ball screw recess **144** and a number of rod recesses **138** to receive a distal end of the ball screw or each rod **132**, respectively. The ball screw recess **144** and rod recesses **138** may extend within the yoke body **143** to accommodate the distal end of the ball screw **116** or each rod **132** (which are positionally fixed), respectively, as the yoke **122** slides along the ball screw **116**. The flange **142** also includes a number of spring recesses **140** to receive a distal end of each spring **130**. Mounting holes **146** are provided around the ball screw recess **144** for securing the ball screw nut **118** to the yoke **122**.

(32) Pin slots **148** extend longitudinally along a length of the yoke body **143** to allow the yoke **122** to slide across the pins **111a**, **111b** extending between the cover plates **112a**, **112b**. A transverse elongated slot **150** is configured to receive a portion of the cam assembly **136**. As the yoke **122** translates along the longitudinal axis **129** of the door operator, the cam assembly **136** rolls or slides along a surface of the slot **150** as the cam assembly **136** pivots about its rotation axis.

(33) FIG. **10** illustrates an example of a cam assembly **136**. First and second cam members **152a**, **152b** are positioned on opposing sides of a guide member **158**. The guide member **158** is configured to engage and ride along the surfaces of the slot **150** of the yoke **122**. The guide member **158** may be fixed in relation to the cam members **152a**, **152b** or may rotate with respect thereto as the guide member **158** rolls along the slot **150**. A spacer **160** may be disposed on either side of the guide member **158** to facilitate rolling of the guide member **158**. A fastener **162** may fasten together the components of the cam assembly **136** and provide an axle on which the guide member **158** can roll.

(34) An output shaft extends from each cam member **152a**, **152b** on either side of the cam assembly **136**. The output shafts **154a**, **154b** extend along the rotation axis **156** of the cam assembly **136**. The first output shaft **154a** may be connected to the spindle **110** for use of the door operator **100** in one installation configuration (e.g., left-handed) and the second output shaft **154b** may be connected to the spindle **110** for use of the door operator **100** in the other installation configuration (e.g., right-handed). Although described as two output shafts, it should be appreciated that the first output shaft **154a** and the second output shaft **154b** may be collectively referred to as an output shaft as they are co-linear and rotate about the same axis.

(35) FIG. **11** provides a perspective view of the door operator **100** with the housing cover **120** and cover plate **112a** removed to avoid obfuscating the components of the drive unit **128**. As will be appreciated by viewing the drive unit in FIG. **11**, operating the motor **114** to draw the ball screw

nut **118** and yoke **122** toward the motor **114** causes the yoke **122** to pull the guide member **158** of the cam assembly **136**. In turn, the cam assembly **136** rotates about the spindle **110**.

(36) FIG. **12** provides a schematic illustration of an embodiment of a door system according to the present disclosure. Actuator **164** may be any suitable device for initiating operation of a swing door operator. Some examples of contemplated actuators include a motion sensor, pressure-activated pad, camera, door-handle, button, remote control, voice-recognition module, RFID reader, keypad, etc. Actuator **164** may transmit an open instruction to a controller associated with motor **114**, thereby causing motor **114** to operate to power the drive unit **128**. The drive unit **128** is operatively connected to both one or more springs **130** and a door **102**. As the drive unit **128** opens the door via powered operation of the motor **114**, the drive unit **128** also compresses the springs **130**. When the controller instructs the motor **114** to shut-off or disengage from the drive unit **128**, the springs **130** expand. Expansion of the springs **130** operates the drive unit **128** in reverse operating mode to swing the door **102** closed.

(37) FIG. **13** provides a flowchart of a method **1000** for operating a door operator such as the door operator **100** in accordance with the present disclosure. At process **1002**, power is provided to the motor causing the motor to rotation the driveshaft. Power may be selectively applied to the motor by a controller. The controller may be in operative communication with an actuator or sensor configured to trigger operation of the door operator. At process **1004**, the ball screw is rotated by the motor. The ball screw may be a portion of the drive shaft of the motor or may be coupled thereto. At process **1006**, a yoke is translated linearly along the ball screw using a ball screw nut secured to the yoke. In some examples, the ball screw nut may be an integral part of the yoke. During translation of the yoke, one or more springs may be compressed between the moving yoke and a stationary part of the door operator. At process **1008**, a cam assembly is rotated by the yoke about an axis of the cam assembly. At process **1010**, rotation of the cam assembly causes rotation of a spindle using an output shaft of the cam assembly. In some examples, the spindle may be an integral part of the cam assembly. At process **1012**, a door is opened using an arm assembly extending between the spindle and the door. At process **1014**, the motor is shut off by stopping the provision of power to the motor. This may be performed by the controller in response to a timer, a door limit sensor, or a position sensor within the door operator (e.g., a rotary encoder tracking a position of the ball screw or motor drive shaft). At process **1016**, the springs may be allowed to expand, releasing the compression and moving the yoke back to its initial position, thereby reversing the movement of the various components caused by operation of the motor.

(38) Persons of ordinary skill in the art will appreciate that the implementations encompassed by the present disclosure are not limited to the particular exemplary implementations described above. In that regard, although illustrative implementations have been shown and described, a wide range of modification, change, combination, and substitution is contemplated in the foregoing disclosure. For example, a number of processes of the method **1000** may be omitted without departing from the scope of the present disclosure. It is understood that such variations may be made to the foregoing without departing from the scope of the present disclosure. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the present disclosure.

Claims

1. A door operator, comprising: an output shaft configured to be coupled to a door; a ball screw; a powered driver operatively coupled to the ball screw and configured to rotate the ball screw in a first rotational direction; and a yoke coupled to the ball screw and configured to translate in response to rotation of the ball screw, the yoke configured to rotate the output shaft, wherein the door operator has: a first installation configuration in which operation of the powered driver to rotate the ball screw in the first rotational direction causes rotation of the output shaft in a first

- output shaft rotational direction; and a second installation configuration in which operation of the powered driver to rotate the ball screw in the first rotational direction causes rotation of the output shaft in a second output shaft rotational direction opposite the first output shaft rotational direction.
2. The door operator of claim 1, further comprising a cam assembly coupled to the yoke, the cam assembly configured to rotate about a rotation axis of the output shaft during translation of the yoke, wherein the rotation axis of the output shaft is substantially transverse to a rotation axis of the ball screw.
 3. The door operator of claim 2, wherein the yoke comprises an elongated slot having a longitudinal axis oriented transverse to the rotation axis of the ball screw, the cam assembly extending into the elongated slot.
 4. The door operator of claim 3, wherein the cam assembly comprises a guide member configured to roll along a surface defining the elongated slot during translation of the yoke.
 5. The door operator of claim 1, further comprising: at least one spring configured to rotate the ball screw in a second rotational direction opposite the first rotational direction.
 6. The door operator of claim 5, further comprising: a ball screw nut disposed about the ball screw and coupled to the yoke, the ball screw nut configured to pull the yoke in a first linear direction when the ball screw is rotated in the first rotational direction and to push the yoke in a second linear direction when the ball screw is rotated in the second rotational direction.
 7. The door operator of claim 5, wherein the yoke is configured to compress the spring as the yoke translates in response to rotation of the ball screw in the first rotational direction, and wherein the spring is configured to expand and push the yoke causing rotation of the ball screw in the second rotational direction.
 8. A system, comprising: a door; and the door operator of claim 1.
 9. The system of claim 8, wherein the door operator further comprises: at least one spring configured to rotate the ball screw in a second rotational direction opposite the first rotational direction.
 10. The system of claim 9, further comprising: a swing arm coupling the door operator and the door.
 11. The system of claim 10, further comprising: a frame hingedly coupled to the door and supporting the door operator.
 12. The system of claim 11, further comprising: a track mounted to the door and slidably supporting an end of the swing arm; or a bracket mounted to the door and pivotally supporting an end of the swing arm.
 13. The system of claim 12, further comprising: an actuator configured to trigger actuation of the powered driver.
 14. A door operator, comprising: an output shaft configured to be coupled to a door; a ball screw; a powered driver operatively coupled to the ball screw and configured to rotate the ball screw in a first rotational direction; a yoke coupled to the ball screw and configured to translate in response to rotation of the ball screw, the yoke configured to rotate the output shaft and having an elongated slot; a cam assembly coupled to the yoke and comprising a guide member extending through the elongated slot; and a directional block configured to prevent the guide member from crossing a longitudinal axis of the door operator.
 15. A door operator, comprising: an output shaft configured to be coupled to a door; a ball screw; a powered driver operatively coupled to the ball screw and configured to rotate the ball screw in a first rotational direction; a yoke coupled to the ball screw and configured to translate in response to rotation of the ball screw, the yoke configured to rotate the output shaft; and at least one pin oriented transverse to a longitudinal axis of the ball screw, the at least one pin configured to extend through at least one slot in the yoke, the at least one slot oriented parallel to the longitudinal axis of the ball screw.
 16. A door operator, comprising: an output shaft configured to be coupled to a door; a ball screw; a

powered driver operatively coupled to the ball screw and configured to rotate the ball screw in a first rotational direction; a yoke coupled to the ball screw and configured to translate in response to rotation of the ball screw, the yoke configured to rotate the output shaft; a spring configured to rotate the ball screw in a second rotational direction opposite the first rotational direction; and a rod extending longitudinally within the spring, wherein the yoke comprises at least one rod recess configured to receive an end of the rod as the yoke is translated.

17. A method for operating a door operator, comprising: rotating a ball screw in a first rotational direction with a powered driver operatively coupled to the ball screw; translating a yoke coupled to the ball screw using the rotation of the ball screw; and rotating an output shaft using the translation of the yoke, the output shaft couplable to a door, wherein the door operator has: a first installation configuration in which operation of the powered driver to rotate the ball screw in the first rotational direction causes rotation of the output shaft in a first output shaft rotational direction; and a second installation configuration in which operation of the powered driver to rotate the ball screw in the first rotational direction causes rotation of the output shaft in a second output shaft rotational direction opposite the first output shaft rotational direction.

18. The method of claim 17, further comprising: compressing, during the translation of the yoke, the at least one spring; and rotating the ball screw in a second rotational direction, opposite the first rotational direction, by expanding the at least one spring.

19. The method of claim 17, wherein rotating the output shaft using the translation of the yoke comprises rotating a cam assembly engaged with the yoke, the cam assembly including the output shaft.
